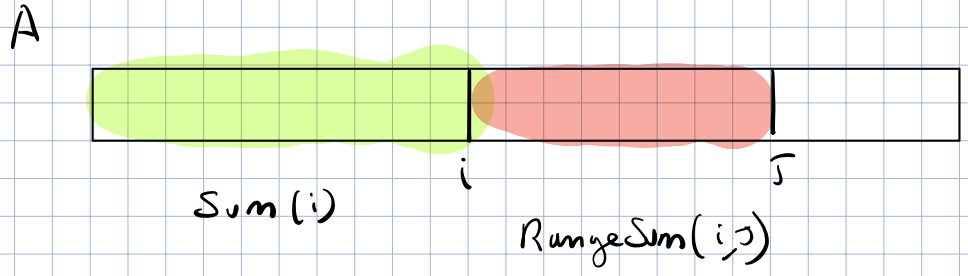
Static Prefix Search

Given $A[1, n]$ of integers, we would like to support:

- $Sum(i) = \sum_{k=1}^{i} A[k]$
- $RangeSum(i, j) = \sum_{k=i}^{j} A[k]$



$$RangeSum(i, j) = Sum(j) - Sum(i-1)$$

Solution

- Store P such that $P[i] = \sum_{k=1}^{i} A[k]$
- $Sum(i)$ takes constant time
- discard A

$$A[i] = RangeSum(i, i+1)$$

- P can be computed in $\Theta(n)$ time

Problem #1

Ilya and queries

Given a binary string $S[1, n]$ consisting of as or bs and a set of m queries

- $q(i, j)$: returns the number of positions $k \in [i, j]$ such that $S[k] == S[k+1]$

Example

P	1	2	3	4	5	6	7
S	a	b	b	a	b	b	a

$q(2, 6) = 2$

Solution in $\Theta(n + m)$ time

P	1	2	3	4	5	6	7
S	a	b	b	a	b	b	a
B	0	1	0	0	1	0	0
P	0	1	1	1	2	2	2

$B[i] = \begin{cases} 0 & \text{if } S[i] \neq S[i+1] \\ 1 & \text{if } S[i] == S[i+1] \end{cases}$

$q(i, j)$ reduces to RunyeSum(i, j)

Problem #2

L. He girl and Queries

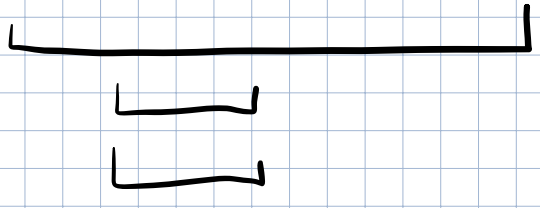
Given $A[1, n]$ and a set of m queries Q.

Each query is a RangeSum(i, j)

Our goal is to permute the values in A to maximize the sum of the results of the queries in Q.

Example

A 5 2 4 1 3



$5 + 6 + 6 = 27$

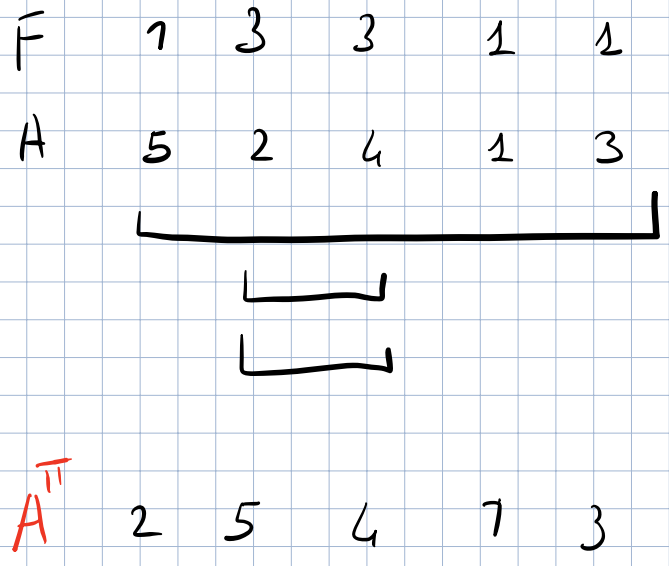
A^{π} 2 5 4 1 3

$5 + 9 + 9 = \boxed{33}$

Solution

- Step #1

  Compute the frequency of each entry



- Step #2

Sort by value and by frequency and match

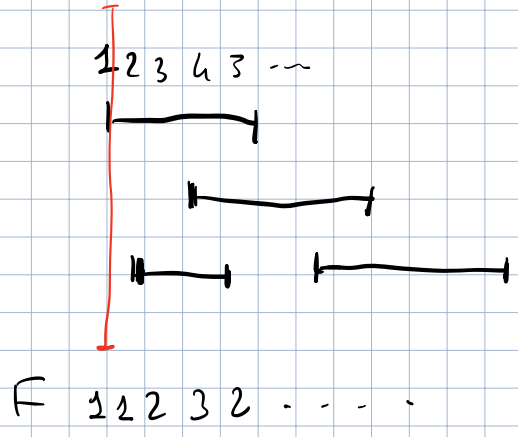
$\Theta(n \log n)$ time

Focus Step #1

Solution #1

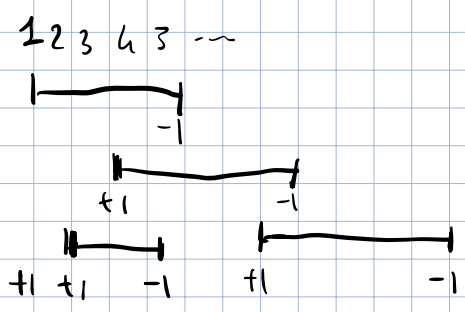
n

Sweep Line



- $\Theta(n + m \log m)$ time with comparison based sorting algorithm

- $\Theta(n+m)$ time with counting sort



C +1 0 +2 -1
1 2 3 4 ...

Sweep line F 1 1 3 2 ...

↖ Prefix Sums of C

q(2,5)

q(3,6)

q(1,2)

C 1 2 3 4 5 6
+1 -1
+1 -1
+1 -1

P 1 2 2 2 2 1

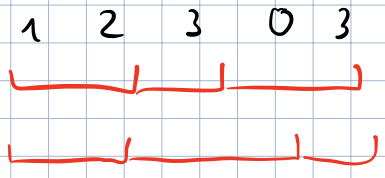
$\Theta(n+m)$ time

Problem #3

Number of Ways

Given $A[1, n]$, count in how many we can split A in 3 parts so that the sum of the parts is the same

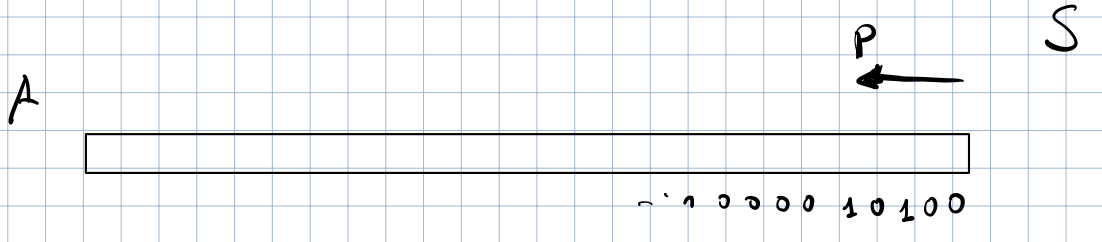
A



(2)

Let's try with the easier problem of splitting in 2 parts

Solution for the easier problem



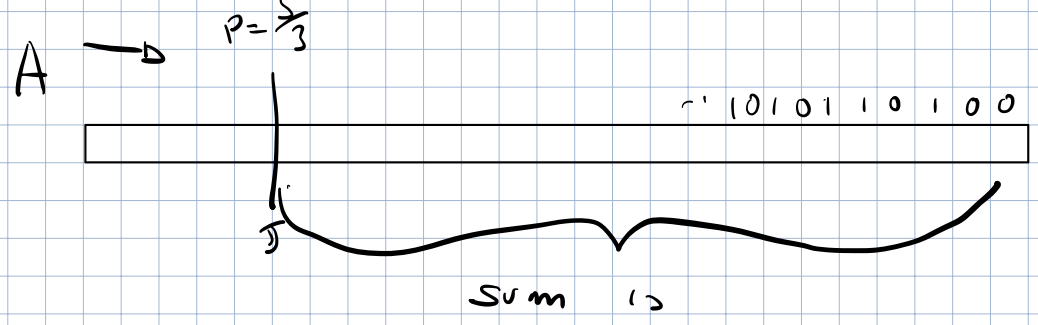
$\Theta(n)$ time

if $P == \frac{S}{2}$ then 1

else 0

Solution for original problem

- if $S \% 3 \neq 0$, no split



sum is

$$2\frac{S}{3}$$

how many ways 2 splitting of this?